

In this Standard the dot has been adopted as the decimal sign

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1. SCOPE

This Standard establishes the requirements for nondestructive inspection (NDI) of internal quality of parts fabricated of composite materials and metal bonded parts.

NOTE

Visual inspection of composite parts shall be performed in accordance with NE 07-009.

2. APPLICABILITY

This Standard is mandatory to parts that are covered by First Part Qualification processes (NE 40-251) conducted in accordance with NE 20-032.

3. APPLICABLE DOCUMENTS

Embraer Standards

NE 01-103 - Parts - Marking Methods.

NE 07-009 - Composite Materials - Visual Inspection.

NE 20-010 - Structural Bonding - Requirements of Parts.

NE 20-032 - Composite Material - Requirements of Parts.

NE 40-251 - Composite Parts, First Part Qualification (FPQ) - Procedures and Requirements.

Foreign Standards

ASTM - American Society for Testing and Materials (USA).

E317 - Standard Practice for Evaluating Performance Characteristics of Ultrasonic Pulse - Echo Testing Instruments and Systems without the Use of Electronic Measurement Instruments.

E1245 - Standard Practice for Determining the Inclusion or Second-Phase Constituent Content of Metals by Automatic Image Analysis.

E1316 - Standard Terminology for Nondestructive Examinations.

E1742 - Standard Practice for Radiographic Examination.

E2491 - Standard Guide for Evaluating Performance Characteristics of Phased-Array Ultrasonic Testing Instruments and Systems.

AIA - Aerospace Industries Association of America (USA)

NAS 410 - Certification and Qualification of Nondestructive Test Personnel.

ASD-STAN - European Association of Aerospace Industries – Standardization

EN 4179 - Qualification and Approval of Personnel for Nondestructive Testing.

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4. DEFINITIONS

The definitions of ASTM E1316 shall apply to terms used in this Standard. The following definitions shall apply to terms which are unknown or have special meaning as used in this Standard.

4.1 ACTIVE APERTURE

Elements of a phased-array probe that are activated as a group to form the ultrasonic beam according to a specific delay law.

4.2 BOND TESTING

A group of NDI techniques that use electronic devices for testing the integrity of an adhesive bond. These techniques are based on mechanical vibrations in the high and low frequency ranges.

4.3 ELEMENTARY PITCH

The distance between the centers of two adjacent elements.

4.4 EMBRAER LEVEL 3

An Embraer's employee certified Level 3 in the applicable method. If a Level 3 in the applicable method is not available, a Level 3 in Ultrasonic Testing shall take the responsibility.

4.5 HIGH FREQUENCY BOND TESTING

Bond testing technique generally operating at frequencies between 100 KHz and 500 KHz. Normally wet couplant is required. Resonance bond testing is included in this category.

4.6 INSPECTION PROCESS

The specific combination of NDI method, NDI technique, personnel qualification, material, equipment, reference standards, and instructions as applied to the inspection of a specific part or group of parts.

4.7 INSPECTION PROCESS, AUXILIARY

An inspection process used to interpret questionable indications when this may not be done solely through the primary inspection process.

4.8 INSPECTION PROCESS, PRIMARY

The inspection process used to detect the probable discontinuities of a specific part or group of parts.

4.9 INSTRUMENTED TAP TESTING

Tap testing technique in which electronic devices are used to tap the area being inspected and/or display results.

4.10 LOW FREQUENCY BOND TESTING

Bond testing technique generally operating at frequencies below 100 KHz. Normally wet coupling is not required. Mechanical Impedance Analysis (MIA) and PITCH-CATCH bond testing are included in this category.

4.11 PASSIVE APERTURE

Element length or probe width of a linear phased-array probe.

4.12 QUESTIONABLE INDICATIONS

Indications of unknown origin.

4.13 REFERENCE SIZE DEFECT (RSD)

A natural or induced/artificial defect that is the same size as the applicable minimum detectable defect size in accordance with NE 20-010 or NE 20-032.

4.14 TAP TESTING

An NDI technique in which vibration is imposed to the part being inspected by a slight tap produced with an appropriate tap device. The discontinuities are revealed by the change in sound they produce. Tapping over good areas normally produces a sharp sound, while tapping over defective areas normally produces a dull sound.

4.15 TIME-ENCODED C-SCAN

C-Scan mapping produced by acquiring data at precise time intervals while the probe is moved without encoding of probe position.

4.16 WRITTEN PROCEDURE

Written description of an inspection process.

4.17 VOID

Air, water vapor, or gas that has been trapped and cured into a laminate or adhesive bond line.

4.18 POROSITY

An internal condition of composite structure which consists of small voids entrapped in a single ply or in the interface between two plies of the laminate.

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5. REQUIREMENTS

5.1 INSPECTION TECHNIQUES

5.1.1 The requirements of this Standard apply to the inspection techniques listed in Table 1.

5.1.2 Use of techniques other than the applicable in Table 1 shall be justified by user and approved by the Embraer Level 3.

TABLE 1 : SELECTION OF NDI TECHNIQUES

NDI Technique	Application						
	LSL	LSL+	LCS	BLB	BMB	BCS	CCS
Ultrasonic Through Transmission	2	1	1	1	1	1	2
Ultrasonic Pulse Echo	1	2	2	2	NA	NA	NA
High Frequency Bond Testing	NA	NA	NA	3	2	NA	NA
Low Frequency Bond Testing	NA	NA	NA	4	3	2	3
Instrumented Tap Testing	NA	NA	NA	NA	NA	3	4
Tap Testing	NA	NA	NA	NA	NA	4	5
Radiography	NA	NA	NA	NA	NA	NA	1

LSL: Laminate defects in composite laminates not bonded to a core and laminate thickness up to 20mm.

LSL+: Laminate defects in composite laminates not bonded to a core and laminate thickness above 20mm.

LCS: Laminate defects in composite laminate skins of cored sandwiches.

BLB: Bondline defects in adhesive composite laminate bonding (composite to composite or composite to metal bonding – except skin to core bonding).

BMB: Bondline defects in adhesive metal bonding (metal to metal bonding – except skin to core bonding).

BCS: Bondline defects in skin to core bonding of cored sandwiches.

CCS: Core defects in cored sandwiches. Applies when NDI for volumetric like defects (core potting voids, lack of foam adhesive, core to edge member voids, excessive core splice gap, etc.) is specifically required.

1, 2, 3, 4, 5: Preference order being 1 the preferred technique and 5 the least preferred technique.

NA: Not Applicable.

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5.2 PERSONNEL QUALIFICATION

Personnel performing tests in accordance with this Standard shall be certified per NAS 410 or EN 4179. Only certified Level 2 or Level 3 individuals are allowed to interpret and evaluate indications for acceptance or rejection of parts and to document test results.

NOTE

Individuals certified in ultrasonic testing and knowledgeable to the principles of the techniques and of the operation of applicable instruments may perform bond testing and instrumented tap testing inspections. Manual tap testing does not require personnel certification.

5.3 MATERIALS

5.3.1 Couplants and additives to couplants, as needed for the applicable inspection technique, shall not be detrimental to parts or subsequent manufacturing operations.

5.3.2 The same couplant or combination of couplant and additives used during system standardization shall be used during inspection.

5.3.3 Only materials listed in Table 2 shall be used as couplants or additives to couplants. Use of materials other than listed in Table 2 shall be justified by user and approved by Embraer Materials Engineering.

TABLE 2 : APPROVED MATERIALS

MATERIAL	FUNCTION	SOURCE	REMARKS
WATER	COUPLANT	ANY AVAILABLE SOURCE	TAP WATER IS ADEQUATE PROVIDED IT IS FREE OF AIR BUBBLES AND CONTAMINANTS THAT MAY INTERFERE WITH INSPECTION RESULTS. USE BIOCIDES OR WETTING AGENT AS NECESSARY.
EXOSSEN No. 30	COUPLANT	K.B. AEROTECH, P.O BOX 350, LEWISTON, PA, USA	(1)
SONOTRACE Gr 20, 30, 40	COUPLANT	MAGNAFLUX 3624 West Lake Avenue, Glenview, IL 60026, USA	(1)
ULTRAGEL II	COUPLANT	MAGNAFLUX 3624 West Lake Avenue, Glenview, IL 60026, USA	DO NOT USE ON MAGNESIUM ALLOYS.
SOUNDCLEAR LV/CF	COUPLANT	DIAGNOSTIC SONAR LIMITED, KIRKTON CAMPUS, LIVINGSTON EH54 7BX	(1)
COUPLANT GEL D	COUPLANT	SOFRANEL 59 rue Parmentier 78500 SARTROUVILLE	(1)
PENETREX	WETTING AGENT	DUBOIS CHEMICALS INCORPORATED 3630 E. Kemper Rd.Cincinnati, OH 45241	-

(1) Water soluble couplants may be used as received or diluted in water, provided the resulting solution is free of air bubbles and contaminants that may interfere with inspection results.

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5.4 EQUIPMENT

5.4.1 Ultrasonic Equipment Requirements

The following apply to ultrasonic flaw detectors, transducers and devices:

a) ultrasonic test instruments shall be conventional or phased-array technology pulse-echo flaw detectors and shall be featured with the following:

- A-Scan display;
- calibrated adjustable gain or attenuation (1 dB steps or finer); and
- pulser/receiver operating in at least the 1 MHz to 10 MHz range.

b) conventional contact transducers for pulse echo applications shall be of the single-element longitudinal-wave type and shall be selected as follows:

- when applicable RSD is 6 mm or less, the element diameter shall not exceed 6 mm (0,25 in);
- when applicable RSD is bigger than 6 mm, the element diameter shall not exceed 13 mm (0,5 in); and
- nominal central frequency shall be 5 MHz or higher.

c) conventional immersion transducers for immersion, bubbler or squirters applications shall be single-element flat or focused type and shall be selected as follows:

- for the inspection of laminates, the nominal central frequency shall be 5 MHz or higher;
- for the inspection of cored sandwiches, the nominal central frequency shall be 1 MHz or higher; and
- water jet diameter shall not exceed 6 mm (0,25 in).

d) phased-array probes shall be selected as follows:

- elementary pitch shall not exceed 1 mm;
- when applicable RSD is 6 mm or less, the passive aperture shall not exceed 7 mm (0,28 in);
- when applicable RSD is bigger than 6 mm, the passive aperture shall not exceed 13 mm (0,5 in); and
- nominal central frequency shall be 5 MHz or higher.

e) ultrasonic instrument and transducer/probe combination shall be capable of:

- consistently and accurately reproducing required indications and responses from known defect inspection standards;
- reliably detecting defects, such as voids, delaminations, inclusions and disbonds of the applicable RSD in parts or structures;
- reliably detecting porosity to the acceptable limits as specified in the applicable material and process specifications; and
- presenting a signal to noise ratio greater or equal to 2 to 1.

f) ultrasonic instruments shall be calibrated in accordance with ASTM E317 or ASTM E2491 as applicable or instrument manufacturer's suggested method. Calibration interval shall be 12 months for analog instruments and 24 months for digital instruments; and

g) transducers that are acceptable for use shall have had their frequency spectrum verified as and compliant to the following:

- band width shall be measured at 6 dB attenuation; and
- actual central frequency deviation from the nominal central frequency shall be within the limits specified in Table 3.

TABLE 3 : REQUIRED TRANSDUCER ACTUAL CENTRAL FREQUENCY

NOMINAL CENTRAL FREQUENCY (NCF)	BAND WIDTH (% of NCF)	ACTUAL CENTRAL FREQUENCY DEVIATION FROM NCF (% of NCF)
NCF $\leq$ 1 MHz	≤ 50	± 20
	> 50	± 30
1 MHz < NCF < 5 MHz	≤ 50	± 15
	> 50	± 25
5 MHz $\leq$ NCF	≤ 50	± 10
	> 50	± 20

5.4.2 Requirements for Bond Testing and Tap Testing Equipment

The following apply to bond and tap testing instruments and tap testing devices:

a) bond testing and instrumented tap testing equipment shall be capable of:

- consistently and accurately reproducing required indications and responses from known defect inspection standards; and
- reliably detecting defects, such as voids, delaminations, inclusions and disbonds of the applicable RSD in parts or structures.

b) bond testing and instrumented tap testing instruments shall be calibrated in accordance with instrument manufacturer's suggested method. Calibration interval shall be 12 months for analog instruments and 24 months for digital instruments; and

c) Figure 1 shows recommended tap testing devices.

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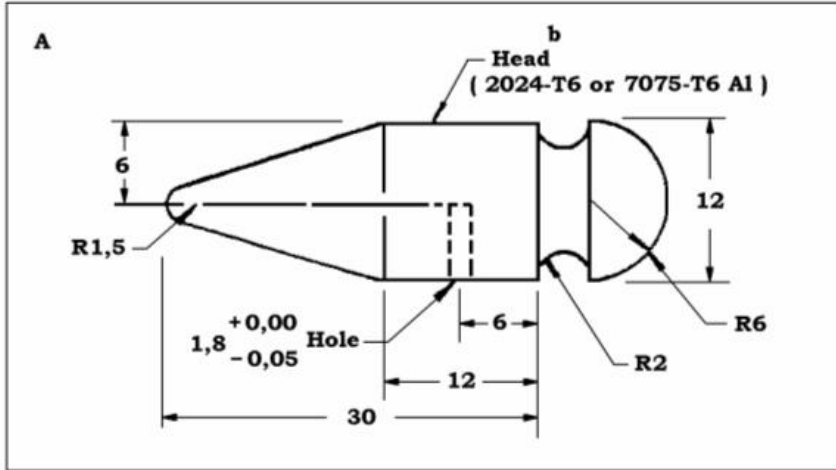
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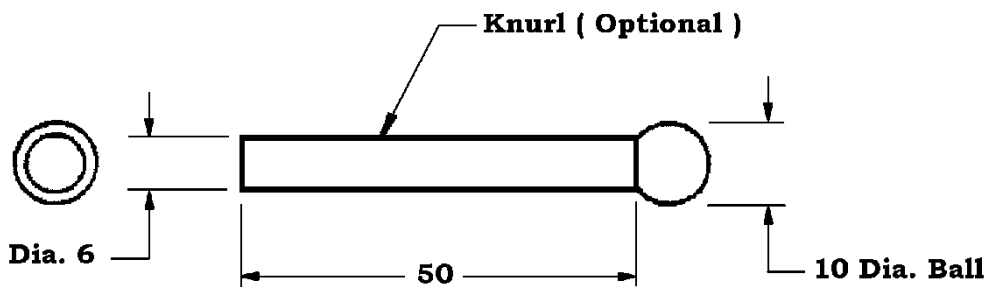
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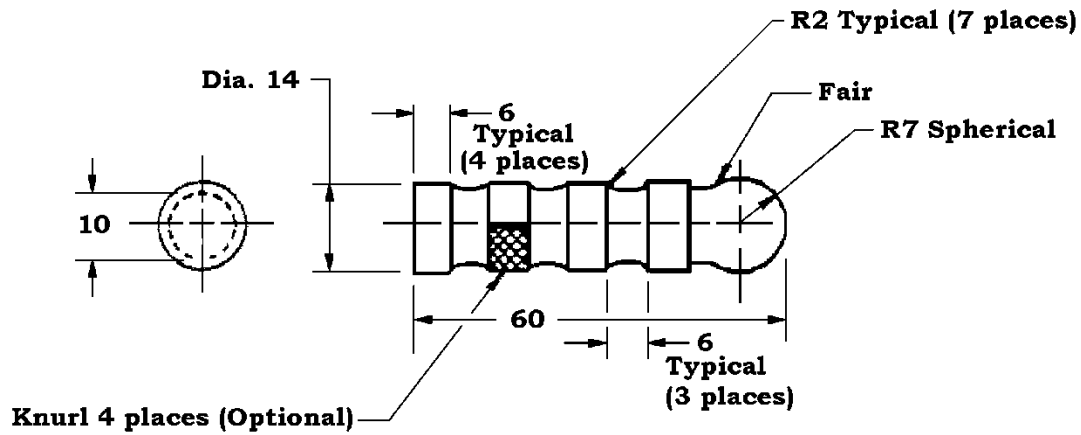


- (**a**) Liquid/paste adhesive may be used if desired. Hole in handle/head may be reduced to provide an interference fit and preclude the need for the adhesive.
- (**b**) Finish 125 Ra: all machined surfaces - handle and stem.
- (**c**) All dimensions are in mm.



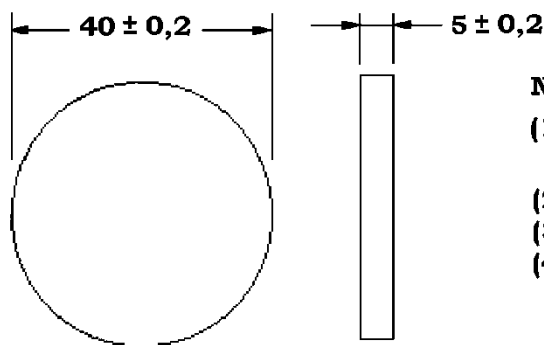
NOTES

- 1) Material - any mild steel or brass bar stock.
- 2) Finish 125 Ra.
- 3) Tolerance $\pm 0,5$.
- 4) Dimensions are in mm.



NOTES

- 1) Material - aluminum 7075-T6 or 2024-T6 bar stock.
- 2) Finish 125 Ra.
- 3) Tolerance $\pm 0,5$.
- 4) Dimensions are in mm.



Notes:

- (1) Material - brass or bronze alloy harder than R B 52.
- (2) Finish 125 Ra.
- (3) Break sharp edges.
- (4) Dimensions are in mm.

FIGURE 1 : TAP TESTING TOOLS

5.4.3 Scanning, Recording and Analysis System

Scanning, recording and analysis system shall comply with the following:

a) unless otherwise approved by the Embraer Level 3 a computer-aided system with C-Scan output shall be used for primary inspection process of all Level 1 parts, regardless the quality class, except the following:

- alternate primary inspection processes may be used to inspect laminate radii smaller than 25 mm; and
- alternate primary inspection processes may be used to inspect up to 0,16 m² of the areas to be inspected of a part.

b) time-encoded C-Scan shall not be used in a primary inspection process;

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- c) for automated scanning systems the scanning index shall not exceed 1/3 of the RSD;
- d) for manual scanning the scanning index shall not exceed 1/2 of the RSD;
- e) for manual scanning the scanning velocity shall not exceed 100 mm/s;
- f) scanning and recording systems shall be capable of producing accurate inspection recordings so that X-Y locations of defects shall be determined on the part within or equal to 15mm;

Note

Scanning and recording systems requirements shall be verified at intervals not exceeding 12 months.

g) when data analysis software is used for defect measurements (length, width, area, attenuation, signal amplitude, etc.) it shall be capable of reliably differentiating between acceptable and unacceptable conditions; and

h) automatic accept-reject decisions without human evaluation are not allowed.

5.4.4 Facilities

Inspection workshops shall comply with the following:

- a) inspection workshops shall be clean and organized in order to provide an environment conducive to undisturbed performance of inspections;
- b) inspection workshops shall be free of sound noise detrimental to satisfactory tap testing inspection;
- c) parts at the inspection workshops shall be protected against humidity, direct sun light and impact damage;
- d) a minimum white light intensity of 1076 lux (100 fc) shall be available at the surface of the part undergoing visual inspection; and
- e) response of equipment to line voltage variations shall be such that changes do not occur in signal amplitude, system effectiveness, or sensitivity for normal line voltage variations. Line voltage regulators external to the equipment shall be used if line voltage fluctuations cause $\pm 5\%$ signal output variation at any instrument power setting.

5.4.5 Radiographic Inspection

The following apply to radiography testing inspection processes:

- a) inspection process shall be in accordance with ASTM E1742; and
- b) inspection process shall be capable of producing radiographs that permit detection of the maximum size defect permitted by referencing specifications, documents or engineering drawings.

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5.4.6 Specific Inspection Tools

Inspection tools required for specific applications such as correction shoes, search unit stand-off attachments, bubblers or water jets shall be detailed or referenced to in the applicable written procedure.

5.5 WRITTEN PROCEDURE

Nondestructive inspection of composite and bonded parts shall be performed in accordance with detailed written procedures.

5.5.1 Written procedures and their revisions shall be approved by a level 3 individual who is certified in the applicable method as per 5.2.

5.5.2 The inspection processes described by the written procedures shall comply with all applicable engineering and specification requirements.

5.5.3 Written procedures may be a single document containing all applicable requirements and instructions, or a combination of a master procedure and technique cards. A master procedure shall contain general information applicable to a group of parts. Technique cards shall provide specific information applicable to the inspection of each specific part.

5.5.4 The following information shall be included, either directly or by reference to the applicable documents:

- a) procedure identification, record of revisions and dates, identification of the individual approving the procedure;
- b) materials and equipment used;
- c) reference standards;
- d) preparation of parts;
- e) detailed instructions for:
 - standardization – setting up of sensitivity, detection of RSD, reference standard to part correction, etc.;
 - inspection – scanning, verification of standardization, identification and mapping of indications, etc.;
 - interpretation of indications – definition of relevance, identification of type of defect, etc.;
 - evaluation of discontinuities in accordance with acceptance criteria; and
 - recording of inspection results.
- f) description of specific inspection tools as per 5.4.6;

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- g) identification of parts (part number or drawing number) to which the written procedure is applicable;
- h) sketch of part or areas where inspection is required;
- i) acceptance criteria; and
- j) inspection process limitations and part specific inspection criteria as per 5.7.6.

5.5.5 Non-production test hardware may be inspected in accordance with a master procedure only.

5.5.6 Written procedures shall be revised whenever any of the following occur:

- a) part design changes affect areas to be inspected in such a way that the inspection process shall be modified or otherwise changed to keep compliance with requirements;
- b) subsequent changes of this or other applicable specifications establish or modify requirements affecting inspection process; or
- c) the inspection process is changed or modified.

5.5.7 When required, suppliers performing inspections according to this specification shall submit their written procedures and revisions thereof for Embraer Level 3 verification.

5.6 COVERAGE AND FREQUENCY OF INSPECTION

Nondestructive inspection according to this Standard shall apply when specifically called on Engineering Drawings.

5.6.1 All parts shall be 100% inspected unless engineering drawings waive inspection of specific part areas.

5.6.2 Unless otherwise specified, inspection is applicable to each production part (sample inspection is not permitted).

5.7 SELECTION OF INSPECTION PROCESSES

The using facility is responsible for establishing of inspection processes compliant to all applicable engineering drawing and specification requirements.

5.7.1 The inspection processes shall be capable of reliably detecting unacceptable flaw conditions defined by engineering drawing and applicable specifications.

5.7.2 The inspection processes shall be consistent with the complexity of the parts in order to assure complete coverage of inspection zones with required sensitivity to likely defects and adequate resolution. It may require, for example, that a combination of techniques be applied to the same inspection zone or that a specific technique be applied to each inspection zone of the same part.

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5.7.3 Inspection techniques shall be selected according to the preference order established in Table 1 or otherwise justified by the user and approved by the Embraer Level 3. Availability of equipment, facilities, personnel and other required means does not justify per se the selection of a specific technique.

5.7.4 Appropriate auxiliary inspection processes shall be established for verification of questionable indications to determine their origin.

5.7.5 Radiographic inspection shall not be used for verification of delaminations or delaminations-like defects.

5.7.6 Part features like small radii and nonparallel surfaces may adversely affect the inspection process capability to completely cover the inspection area or to detect and characterize a specific type of defect accordingly. When the applicable primary and auxiliary inspection processes are not able to overcome such limitations, a part specific inspection criteria compliant to the applicable part acceptance criteria shall be established by the user and approved by the Embraer Level 3. Figure 2 shows examples of part features that may adversely affect the inspection process.

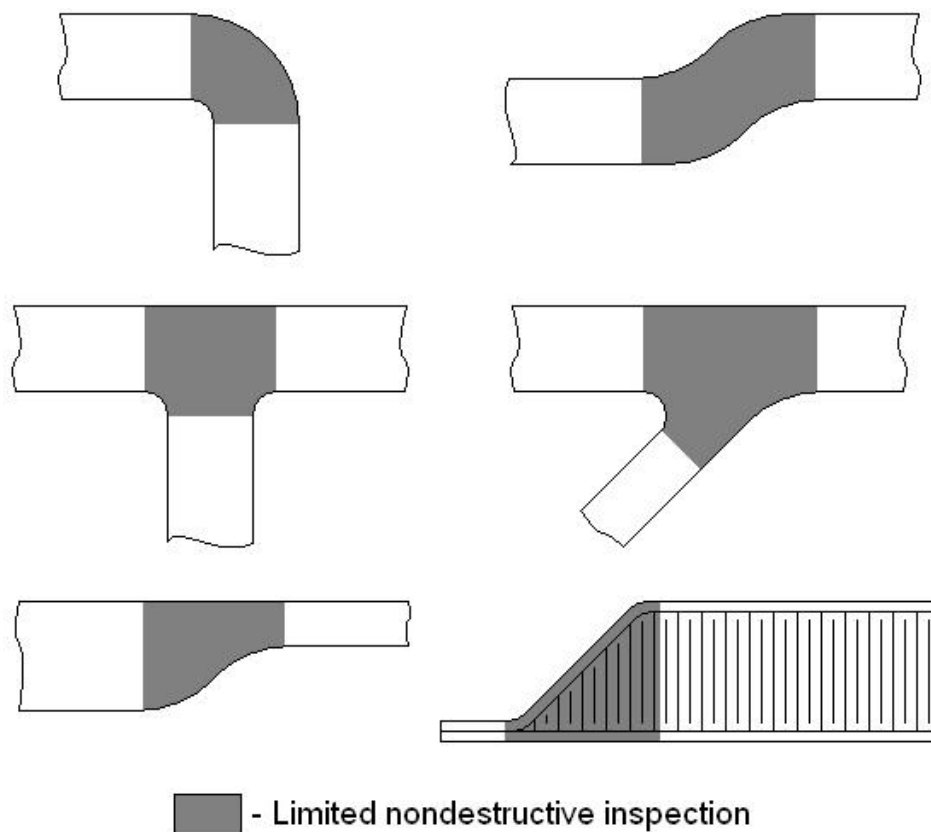


FIGURE 2 : PART FEATURES THAT MAY ADVERSELY AFFECT THE INSPECTION PROCESS

5.8 REFERENCE STANDARDS

Reference standards are required to standardize equipment response to known conditions providing a comparison base for acceptance or rejection of inspected products.

5.8.1 General Requirements

Reference standards shall represent configuration and complexity of parts being inspected with respect to material types, shape, skin or laminate thickness, adhesive or bonding medium, internal structure, and fabrication processes and shall comply with the following:

- a) unless otherwise specified, reference standards shall be used for all acceptance inspections;
- b) reference standards applicable to ultrasonic inspection of composite laminates shall have laminate thickness within $\pm 0,5$ mm of the laminate thickness of the part being inspected;
- c) when ultrasonic through-transmission is used, the core height of cored sandwich reference standards shall be within $\pm 25\%$ of that of the applicable part;
- d) reference standards shall contain RSD adequately located with respect to part configuration and complexity. Except for RSD, standards shall be free of flaws;
- e) the position and size of RSD shall be verified and recorded by appropriate means like C-Scan maps or radiographs. Data sample size of C-Scan maps shall be 1,0 mm x 1,0 mm or smaller;
- f) Figures 8, 9 and 10 show recommended methods for preparation of RSD;
- g) for pulse-echo applications, the signal from each RSD shall achieve at least the amplitude specified in Table 4 when compared to the backwall echo at the same depth ($\pm 0,5$ mm);
- h) for through-transmission applications, the attenuation of the transmission signal at each RSD shall be at least 10 dB when compared to the best transmission signal from surrounding areas;
- i) reference standard composite laminates shall be tested for fiber and resin content according to material specification (MEP), and thickness per ply shall be within the acceptance criteria defined in NE 20-032;
- j) reference standard composite laminates shall be tested for Barcol hardness in accordance with NE 40-251;
- k) reference standard composite laminates shall be tested for porosity content in accordance with NE 40-251 or by image analysis methods in accordance with ASTM E1245;
- l) each reference standard shall be given a permanent unique identification;

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m) records of compliance to the applicable requirements shall be kept for each reference standard; and

n) use of reference standards other than herein specified shall be justified by the user and approved by the Embraer Level 3.

TABLE 4 : MINIMUM RSD ECHO SIGNAL AMPLITUDE

Transducer element diameter	Backwall echo amplitude	Class 1 RSD amplitude	Class 2 RSD amplitude	Class 3 & 4 RSD amplitude
6 mm	100%	90%	90%	90%
0,25 in	100%	80%	90%	90%
10 mm	100%	NA	90%	90%
0,5 in	100%	NA	60%	90%

5.8.2 Material Reference Standards (MRS)

The MRS are designed to represent the material of the bulk of composite laminates. MRS as specified herein shall be used for inspection of laminates with surface curvature radii equal to or bigger than 25 mm. They may be used for inspection of smaller radii provided a correction factor is established and justified through the qualification of the inspection process as per 6.1.

The MRS shall comply with the following:

a) laminate materials shall be of the same type as those of the applicable parts;

NOTE

Same type means that both MRS laminate and inspected part have the same type of materials: fiber, resin, sizing, weave style; and the same characteristics such as thickness per ply.

b) fabrication process parameters shall be adjusted in order to produce a high quality laminate with low void content;

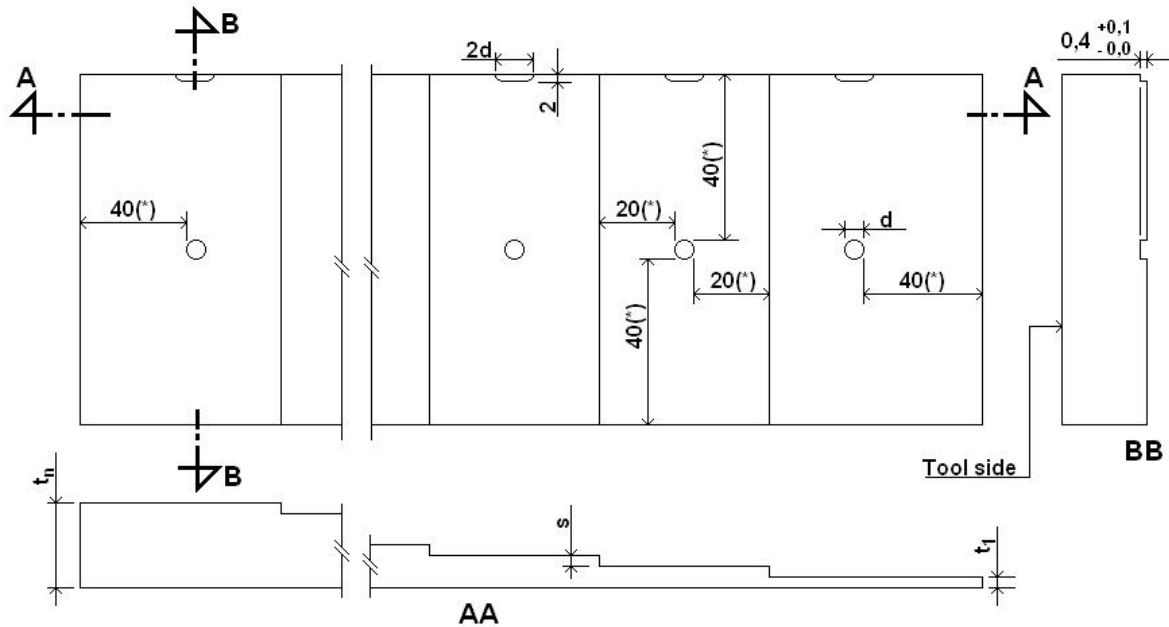
c) ply sequence and orientation shall be balanced in order to prevent warpage;

d) fiber to resin ratio shall comply with the acceptance criteria defined in material specification (MEP), and thickness per ply shall be within the acceptance limits, in accordance with NE 20-032;

e) Barcol hardness shall be within acceptance limits, in accordance with NE 40-251;

f) porosity content shall not exceed 0,2 percent; and

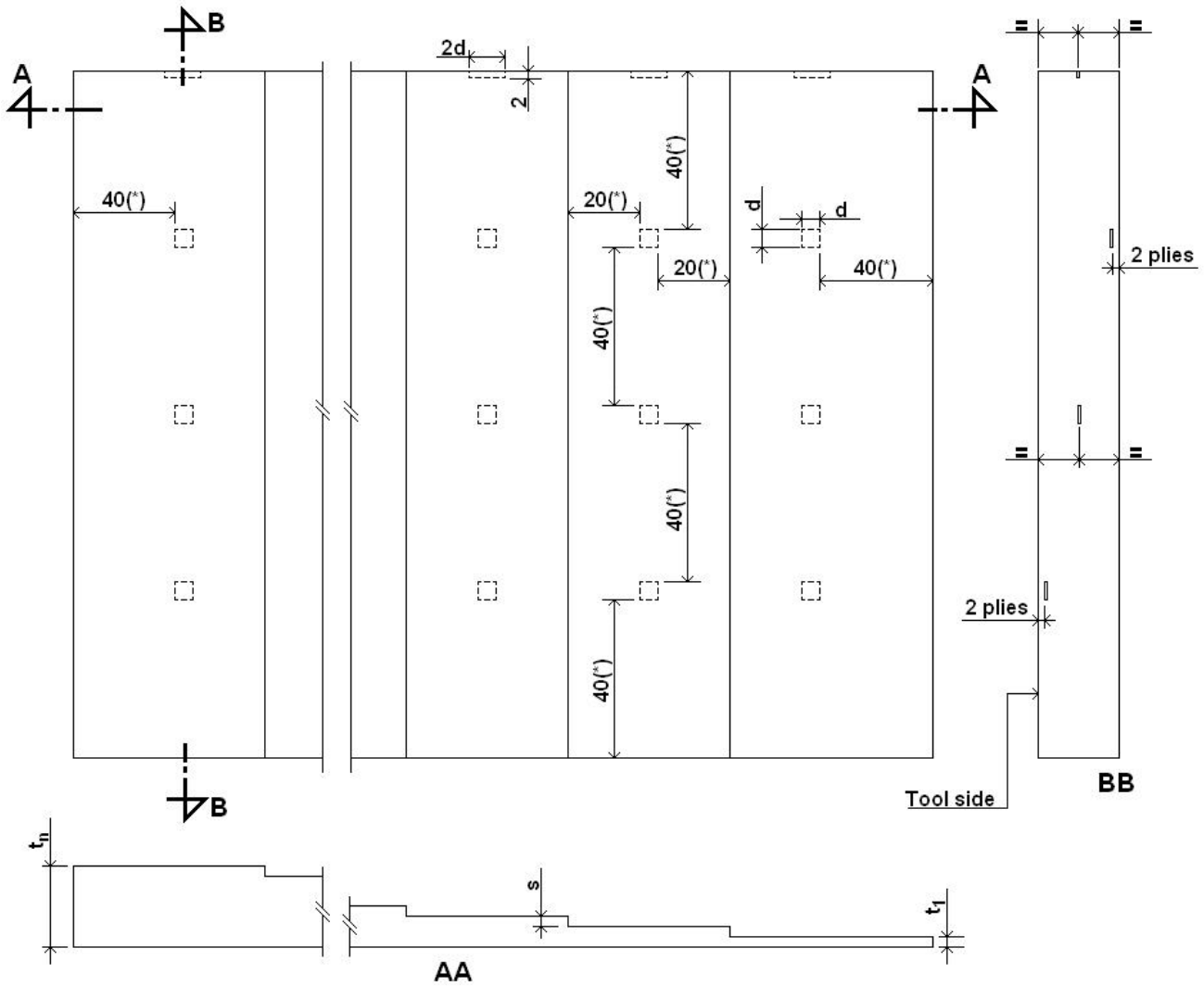
g) MRS shall be fabricated in accordance with Figures 3 or 4.



- d – Diameter of the RSD in accordance with the applicable inspection class. All RSD shall be unplugged flat bottom holes and slots 0,4 mm to 0,5 mm deep.
- t_1 – Thickness of the first step = 4 plies or 1 mm, whichever is less.
- t_n – Thickness of the last step $\geq$ applicable part laminate thickness.
- s – Step thickness increment = 4 plies or 1 mm, whichever is less.
- (*) Minimum distance from the edges of the RSD.

The applicable laminate thickness range may be covered by a single reference standard or by a set of shorter ones provided the step thickness increment is kept compliant to requirements.

FIGURE 3 : MRS FOR PULSE-ECHO APPLICATIONS



d – Side of squared inserts (or diameter of round inserts) = diameter of the RSD in accordance with the applicable inspection class.

All RSD and edge inserts shall be embedded inserts prepared as per 5.8.1.

Place RSD 2 plies from front and back surfaces and at the mid-laminate thickness. Place edge inserts at the mid-laminate thickness.

t₁ – Thickness of the first step = 4 plies or 1 mm, whichever is less.

t_n – Thickness of the last step ≥ applicable part laminate thickness.

s – Step thickness increment = 4 plies or 1 mm, whichever is less.

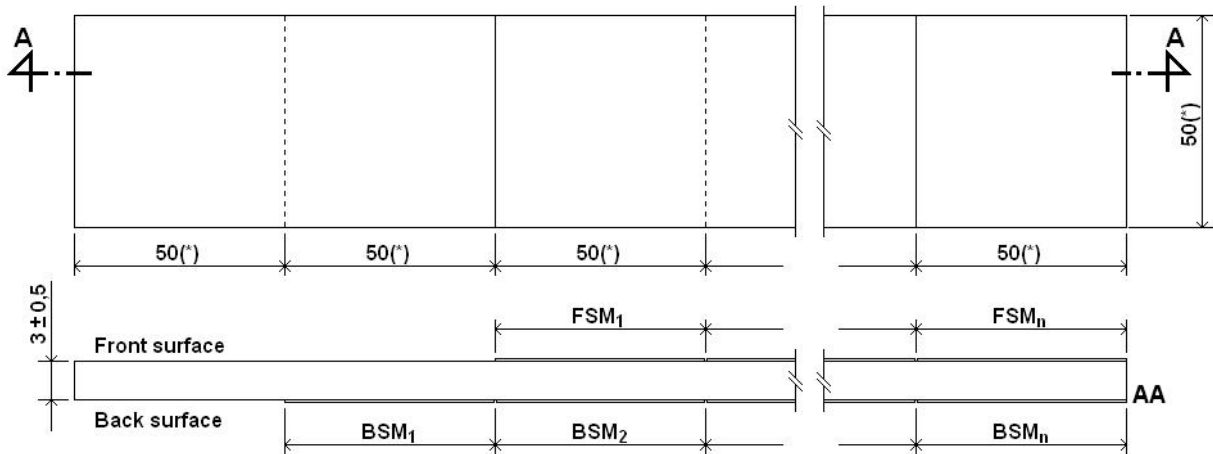
(*) Minimum distance from the edges of the RSD.

FIGURE 4 : MRS FOR PULSE-ECHO OR THROUGH-TRANSMISSION APPLICATIONS

5.8.3 Surface Reference Standards (SRS):

The SRS are designed to represent the materials of the front and back surfaces of composite laminates (copper mesh, PVF film, fiberglass, etc.) and shall comply with the following:

- surface materials shall be the same as those of the applicable parts;
- substrate materials shall be of the same type as those of the applicable parts;
- ply sequence and orientation shall be balanced in order to prevent warpage;
- fiber to resin ratio and porosity content shall comply with the acceptance criteria defined in material specification (MEP), and thickness per ply shall be within the acceptance limits, in accordance with NE 20-032;
- Barcol hardness shall be within acceptance limits, in accordance with NE 40-251; and
- SRS shall be fabricated in accordance with Figure 5.



FSM₁, ..., FSM_n – Front surface materials.

BSM₁, BSM₂, ..., BSM_n – Back surface materials.

(*) Minimum patch size (50 mm x 50 mm).

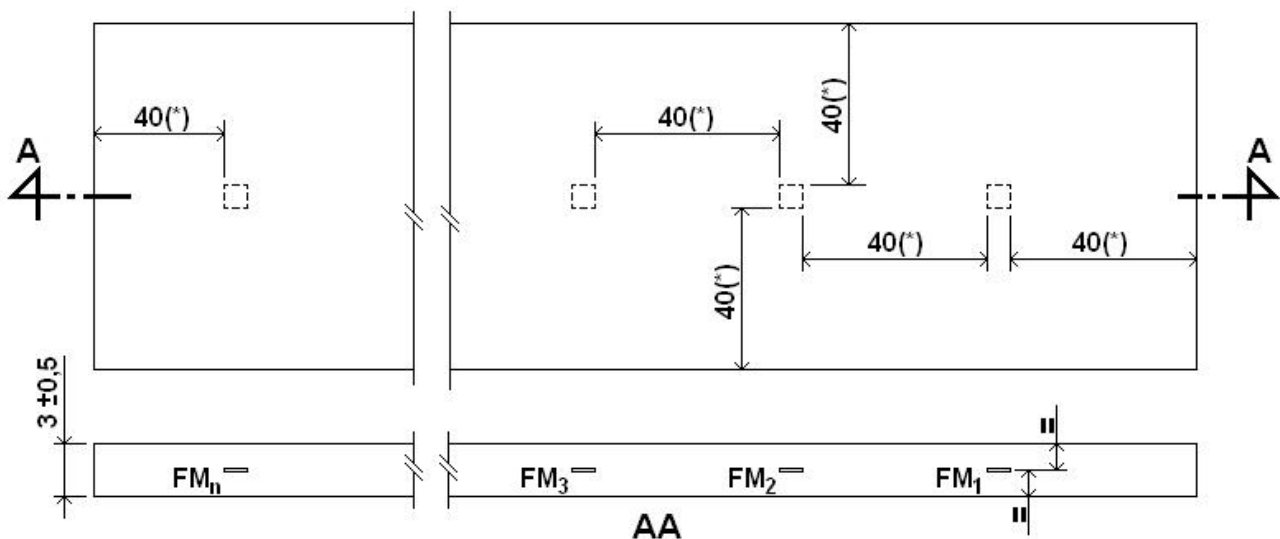
The applicable combinations of front and back surface materials may be covered by a single reference standard or by a set of separate SRS provided an area of bare substrate front and back surfaces is included in each SRS.

FIGURE 5 : SURFACE REFERENCE STANDARD

5.8.4 Foreign Material Reference Standards (FMRS)

The FMRS are designed to represent inclusions of foreign materials that may be inadvertently incorporated inside the laminate and shall comply with the following:

- a) every material that is found within the environment of lamination of parts and that may be inadvertently incorporated inside the laminate shall be included as an inclusion in the FMRS;
- b) laminate bulk materials shall be of the same type as those of the applicable parts;
- c) ply sequence and orientation shall be balanced in order to prevent warpage;
- d) fiber to resin ratio and porosity content shall comply with the acceptance criteria defined in material specification (MEP), and thickness per ply shall be within acceptance limits, in accordance with NE 20-032;
- e) Barcol hardness shall be within acceptance limits, in accordance with NE 40-251; and
- f) FMRS shall be fabricated in accordance with Figure 6.



FM₁, ..., FM_n – Foreign materials inserts. All inserts shall be one single ply square of each applicable material. The size of inserts shall be the maximum allowable for internal inclusions in accordance with the applicable inspection class.

(*) Minimum distance from the edges of the inserts.

The applicable foreign material inclusions may be covered by a single reference standard or by a set of separate FMRS.

FIGURE 6 : FOREIGN MATERIAL REFERENCE STANDARD

5.8.5 Honeycomb Cored Reference Standards (HCRS)

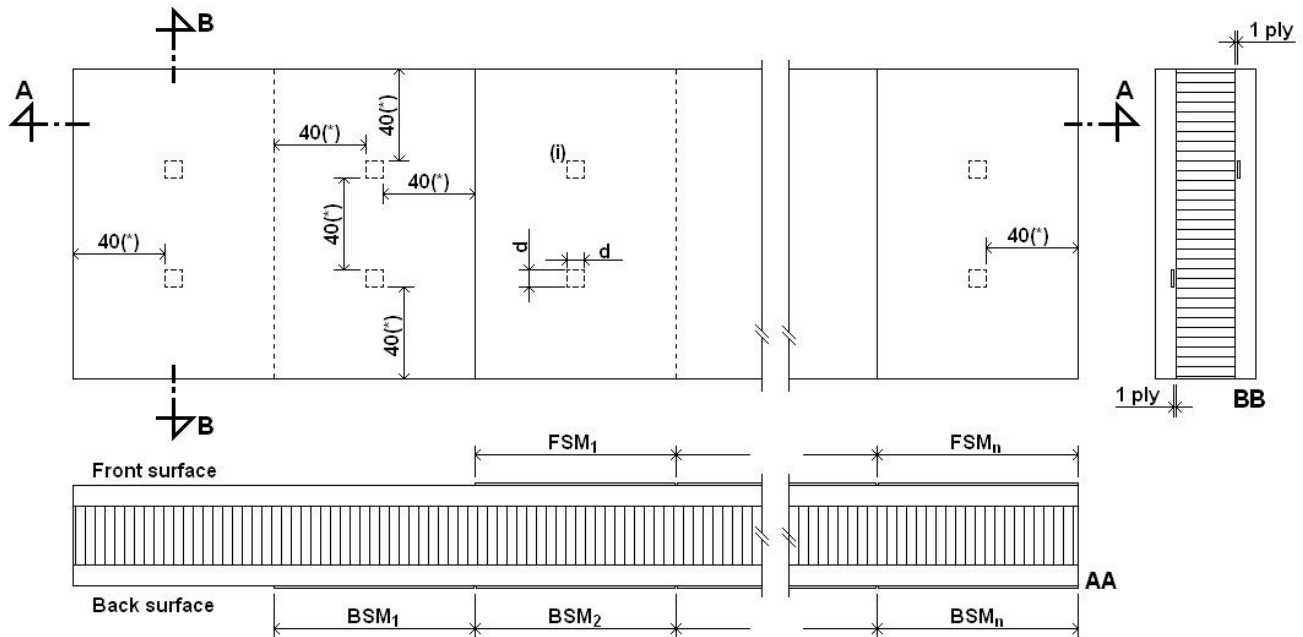
The HCRS are designed to represent sandwich panels made of honeycomb core and composite laminate skins and shall comply with the following:

- a) laminate skin bulk, skin surface, adhesive and core materials shall be of the same type as those of the applicable parts;
- b) HCRS shall be fabricated by secondary bonding (laminate skins shall be cured before bonding to the honeycomb core);
- c) fabrication process parameters shall be adjusted in order to produce a high quality laminate with low void content;
- d) ply sequence and orientation shall be balanced in order to prevent warpage;
- e) fiber to resin ratio shall comply with the acceptance criteria defined in material specification (MEP), thickness per ply shall be within acceptance limits, in accordance with NE 20-032;
- f) Barcol hardness shall be within acceptance limits, in accordance with NE 40-251;
- g) porosity content shall not exceed 0,2 percent; and
- h) HCRS shall be fabricated in accordance with Figure 7.

5.8.6 Part Specific Reference Standards (PSRS)

The PSRS are designed to represent specific parts or part groups that due to their shape or configuration shall not be represented by flat MRS as per 5.8.2. They are also designed to represent those part features that may adversely affect the inspection process as per 5.7.6. PSRS shall comply with the following:

- a) materials and fabrication process parameters shall be the same as those of the parts they represent (see Figures 8, 9 and 10);
- b) PSRS shall have applicable RSD placed at the full range of possible discrepancy locations such as flat areas, radii, bondlines and those part features as per 5.7.6;
- c) fiber to resin ratio and porosity content shall comply with the acceptance criteria defined in material specification (MEP), and thickness per ply shall be within acceptance limits, in accordance with NE 20-032;
- d) Barcol hardness shall be within acceptance limits, in accordance with NE 40-251; and
- e) PSRS shall be justified by the user and approved by the Embraer Level 3.



d – Side of squared inserts (or diameter of round inserts) = diameter of the RSD in accordance with the applicable inspection class.

All RSD shall be embedded inserts prepared as per 5.8.1.

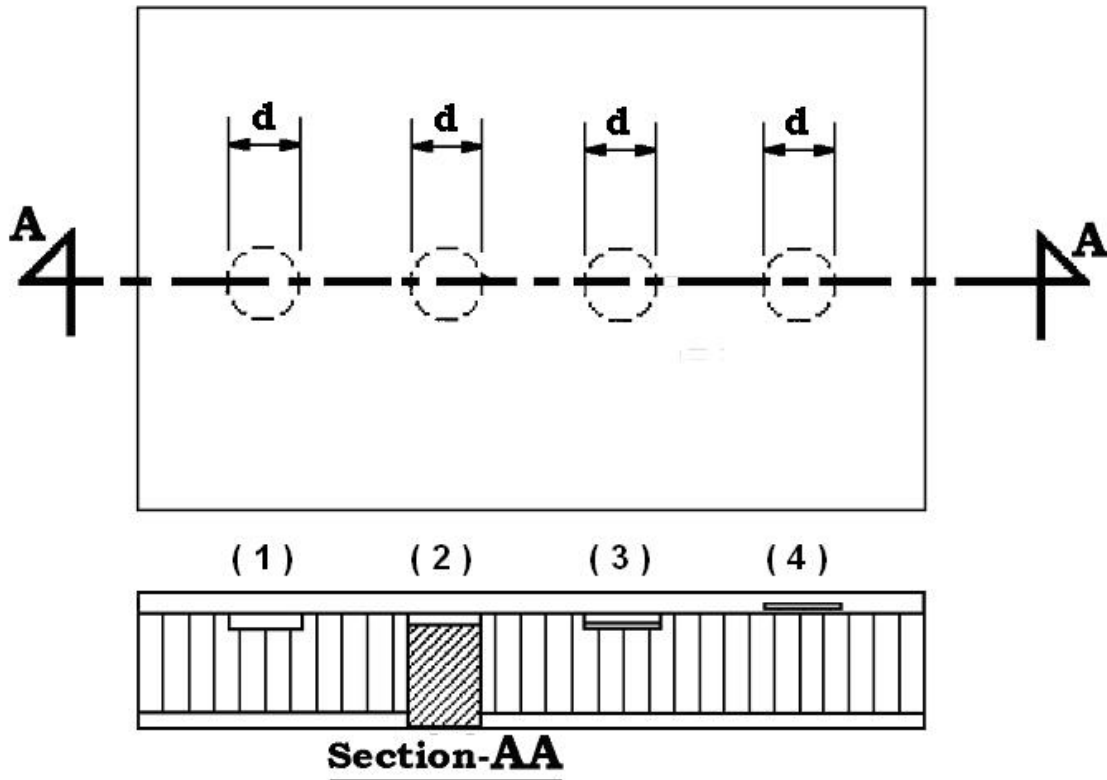
FSM1, ..., FSMn – Front surface materials.

BSM1, BSM2, ..., BSMn – Back surface materials.

(*) Minimum distance from the edges of the RSD.

The applicable combinations of front and back surface materials may be covered by a single reference standard or by a set of separate HCRS.

FIGURE 7 : HONEYCOMB CORED REFERENCE STANDARD



(1) Cell walls broken 3 mm down.

(2) Plugged flat bottom hole. When plugging flat bottom holes, do not allow adhesive on ends. Apply adhesive to circumference only. Plugging is not required for single side NDI techniques, e.g., low frequency bond testing.

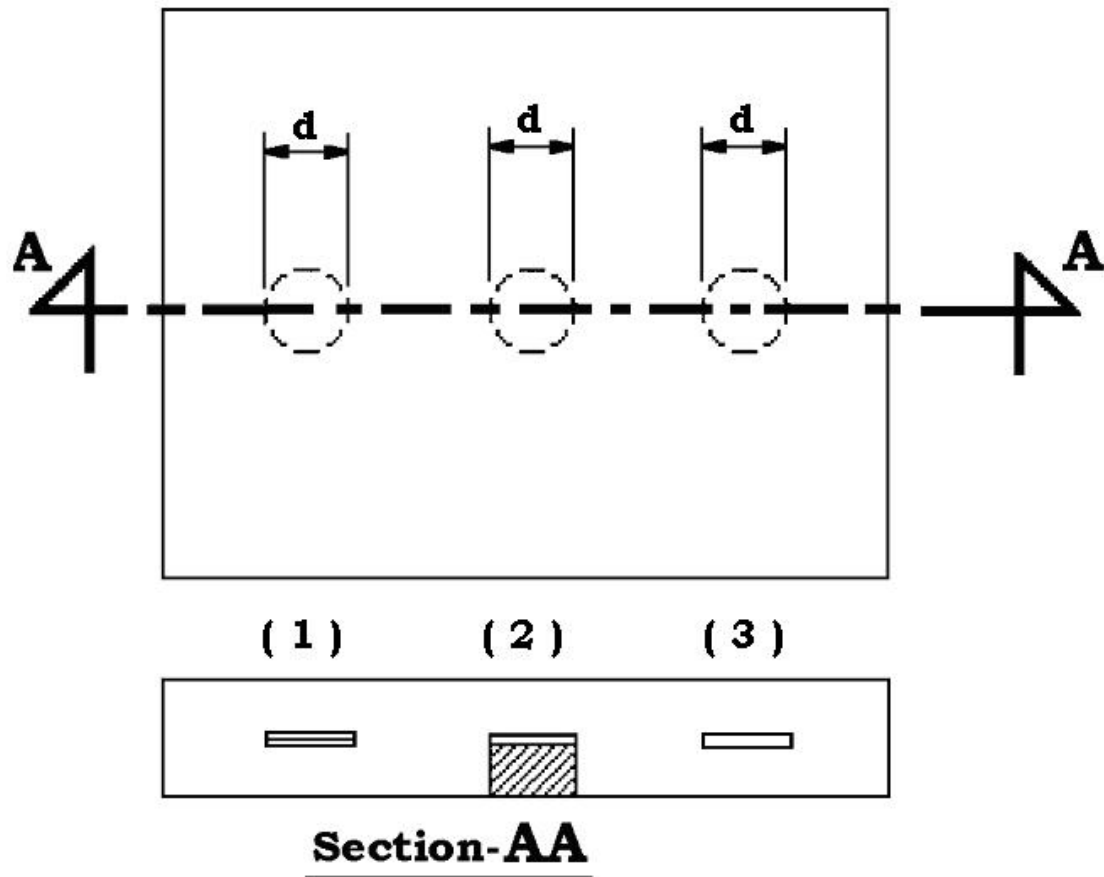
(3) Two overlapped 0,15 mm MAX gage Teflon inserts (round or squared). Care shall be taken to prevent slippage of the Teflon inserts with respect to each other during lay up or cure. The response obtained from the inserts shall be clearly distinguishable from that of good areas.

(4) Metal foil (copper, brass or steel) round or squared inserts 0,20 mm to 0,25 mm thick. Coat inserts with a release agent and place them 1 laminate ply from the core.

d – Diameter of the RSD in accordance with the applicable inspection class.

Seal open edges of honeycomb.

FIGURE 8 : PREPARATION OF RSD IN HONEYCOMB CORED PANELS



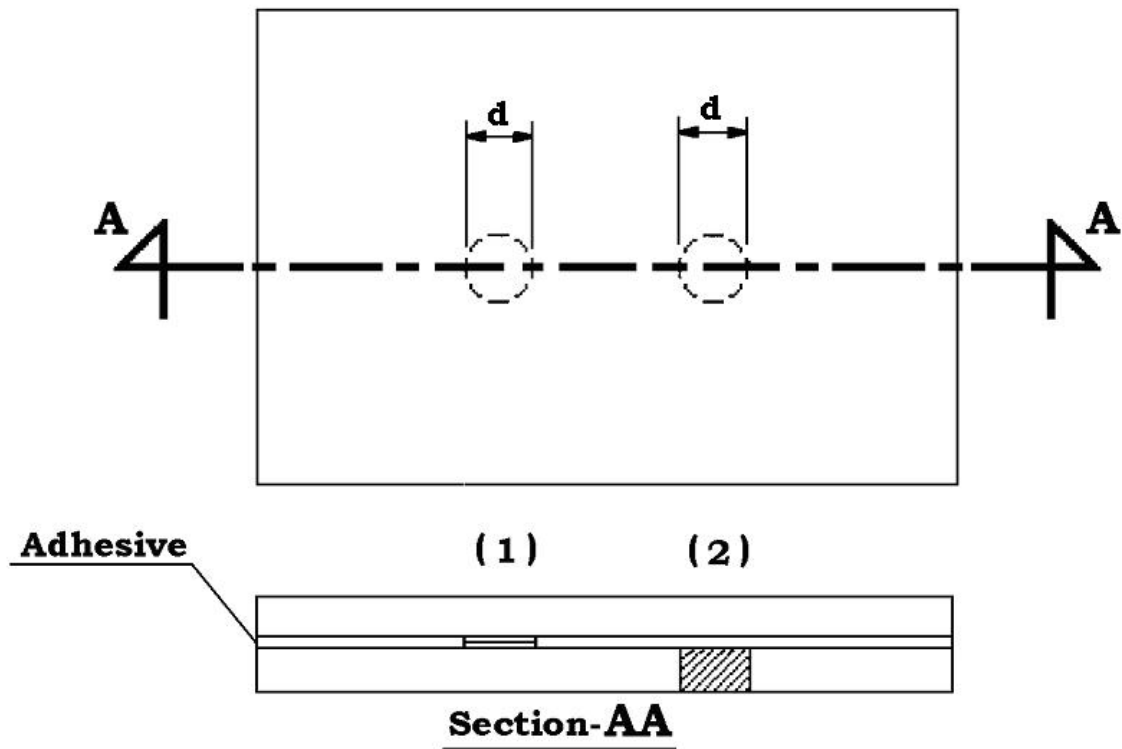
(1) Two overlapped 0,15 mm MAX gage Teflon inserts (round or squared). Care shall be taken to prevent slippage of the Teflon inserts with respect to each other during lay up or cure. The response obtained from the inserts shall be clearly distinguishable from that of good areas.

(2) Plugged flat bottom hole. When plugging flat bottom holes, do not allow adhesive on ends. Apply adhesive to circumference only. Plugging is not required for single side NDI techniques, e.g., ultrasonic pulse-echo.

(3) Metal foil (copper, brass or steel) round or squared inserts 0,20 mm to 0,25 mm thick. Coat inserts with a release agent.

d – Diameter of the RSD in accordance with the applicable inspection class.

FIGURE 9 : PREPARATION OF RSD IN COMPOSITE LAMINATES



(1) Two overlapped 0,06 mm MAX gage Teflon inserts (round or squared). Care shall be taken to prevent slippage of the Teflon inserts with respect to each other during lay up or cure. The response obtained from the inserts shall be clearly distinguishable from that of good areas.

(2) Plugged flat bottom hole. When plugging flat bottom holes, do not allow adhesive on ends. Apply adhesive to circumference only. Plugging is not required for single side NDI techniques, e.g., high frequency bond testing.

d – Diameter of the RSD in accordance with the applicable inspection class.

FIGURE 10 : PREPARATION OF RSD IN BONDED STRUCTURES

5.8.7 Combination of Reference Standards

Provided all applicable requirements are met, the design of MRS, SRS, FMRS and HCRS may be modified to make combined reference standards or to include RSD of different applicable inspection classes.

5.8.8 Validation of Reference Standards at the Embraer

Reference standards in use at the Embraer shall be validated as follows:

- a) the validation shall be performed by the Embraer Level 3;
- b) a report of validation shall be issued for each reference standard and shall include:
 - identification;
 - materials;
 - fabrication process;
 - drawing or sketch showing location and size of RSD;
 - applicable parts or groups of parts;
 - justification of use as per 5.8.1.p) when applicable; and
 - identification and signature of the responsible for the validation.
- c) the validation report shall be revised whenever the list of applicable parts or groups of parts is modified.

5.9 EQUIPMENT STANDARDIZATION

The equipment shall be standardized, using the applicable reference standard(s).

5.9.1 When ultrasonic phased-array is used the active aperture shall be set to be approximately the same size as the passive aperture.

5.9.2 Equipment sensitivity shall be set using the applicable MRS, HCRS or PSRS to establish signal levels that ensure the following:

- a) detection of applicable RSD;
- b) detection of unacceptable porosity content – when required by the applicable part acceptance criteria;
- c) detection of inserts of the applicable FMRS; and
- d) sizing of RSD within 100% to 125% of actual size.

5.9.3 When part surface materials differ from those of the applicable MRS, the sensitivity shall be corrected using the applicable SRS.

5.9.4 Equipment standardization shall be performed prior to beginning of inspection.

5.9.5 Standardization shall be checked periodically during inspection and at the end of it. If abnormalities are observed in the equipment operation or the equipment found out of standardization, all material tested subsequent the last acceptable standardization.

5.9.6 When ultrasonic through transmission techniques are used, the amplitude of the best transmission signal on the part shall be within ± 2 dB of that signal on the applicable MRS and within ± 3 dB of that signal on the applicable HCRS.

5.9.7 When ultrasonic pulse-echo techniques are used, the amplitude of the best back reflection on the part shall be within ± 3 dB of that signal on the applicable MRS.

5.10 PARTS PREPARATION

Parts shall be clean and free of any surface conditions detrimental to satisfactory inspection.

5.10.1 Prior to nondestructive inspection, the parts shall be visually examined for surface and edge conditions.

5.10.2 Surface conditions that may affect inspection results shall be removed whenever possible. When surface conditions preclude adequate disclosure and evaluation of discontinuities the affected parts shall be rejected.

5.10.3 When applicable delaminated edges and exposed openings shall be sealed to prevent ingress of couplant that may interfere with inspection results or be retained inside the parts.

5.11 INSPECTION, INTERPRETATION AND EVALUATION

5.11.1 The conditions established during equipment standardization (sensitivity, equipment adjustments, couplants, index, scanning rate, etc.) shall be kept during inspection of parts.

5.11.2 Radiographic inspection shall be performed in accordance with ASTM E1742.

5.11.3 All indications that are equal to or greater than the RSD shall be interpreted.

5.11.4 Interpretation of indications shall determine the type, size and depth of discontinuities as required for defect evaluation according to the applicable acceptance criteria.

5.11.5 The following conditions shall be considered indications:

- a) any signal reaching the signal levels established in accordance with 5.9.2;
- b) an attenuation of transmission signal equal to or greater than 6 dB – when ultrasonic through-transmission techniques are used;
- c) an attenuation of the back reflection equal to or greater than 12 dB when ultrasonic pulso-echo techniques are used;
- d) an intermediate signal with amplitude equal to or greater than 50% of the amplitude of the applicable RSD at the same depth ($\pm 0,5$ mm) – when ultrasonic pulse-echo techniques are used; and
- e) other conditions as specified in the written procedure.

5.11.6 Questionable indications shall be verified with appropriate auxiliary inspection processes to determine their cause.

- a) When questionable indications are not proved as being non relevant, they shall be evaluated for acceptance on the basis of primary inspection process results.
- b) When both sides of the laminate are accessible indications of delaminations shall be verified from both sides for hidden defects.
- c) Indications of accumulated porosity bigger than the applicable RSD shall be verified for delaminations or voids.

5.11.7 Unless otherwise specified unacceptable porosity content in composite laminates (single laminates, bonded laminates, laminate skins of cored sandwiches) shall be indicated by the signal attenuation established in Table 5.

TABLE 5 : ATTENUATION LEVEL FOR POROSITY INDICATION

LAMINATE THICKNESS t (mm)	SIGNAL ATTENUATION (dB)				
	THROUGH TRANSMISSION			PULSE ECHO	
	HCRS	MRS	PSRS (1)	MRS	PSRS (1)
$t \leq 2$	9	6	6	12	12
$2 < t \leq 5$	10	7	6	14	12
$5 < t \leq 9$	11	8	6	16	12
$9 < t \leq 12$	12	9	6	18	12
$12 < t$	13	10	6	20	12

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continued

t – laminate thickness

The signal attenuation in this Table is:

in through transmission – the difference between the transmission signal through the defective area and the transmission signal through a comparable area of the applicable reference standard; or

in pulse-echo – the difference between the back reflection from the defective area and the back reflection from a comparable area of the applicable reference standard.

Unless the affected laminate be identified by the applicable inspection process, “t” is the thickness of the thinnest laminate in a multi-laminate configuration, e.g., bonded laminates or laminate skins of cored sandwiches.

(1) Applicable to part shape, configuration or features as per 5.8.6.

5.11.8 Unacceptable discontinuities shall be recorded in such a way they may be specifically located on the part.

5.11.9 Where required, locate all unacceptable discontinuities on the part with a suitable marking device that is not detrimental to the part according to NE 01-103.

5.12 ACCEPTANCE CRITERIA

Acceptance criteria for defect evaluation shall be as established by engineering drawings or applicable specifications.

5.1.2.1 Unless otherwise specified composite parts shall be evaluated in accordance with NE 20-032.

5.12.2 Unless otherwise specified metal bonded parts shall be evaluated in accordance with NE 20-010.

5.13 INSPECTION RECORDS

5.13.1 Unless otherwise specified in contract, inspection records shall be maintained in compliance with the Quality Assurance documents.

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5.13.2 Inspection records shall include the following as applicable:

- a) part or assembly number;
- b) quantity of parts presented for acceptance testing;
- c) quantity of parts tested;
- d) disposition of parts (accept/reject);
- e) date of inspection;
- f) NDI report or rejection report number, when required;
- g) unacceptable discontinuities description including type of defect, location, size and depth as applicable;
- h) C-Scan files of Level I parts; and
- i) work order or part serial number.

5.13.3 A monthly rejection rate of parts inspected in the last 12 months shall be kept available for verification upon request.

5.14 MARKING

5.14.1 Parts inspected according to this Standard and found to conform to the requirements of the applicable acceptance criteria shall be marked as follows:

- a) when parts are permanently identified or are too small to be marked, the records accompanying the parts shall be marked; and
- b) when parts are not permanently identified, the parts shall be marked as appropriate according to NE 01-103. The records accompanying the parts shall be marked also.

5.14.2 Acceptance marking shall identify the inspection facility, the type of inspection and the inspector.

5.14.3 Parts containing discontinuities exceeding the acceptance limits shall be identified as defective, segregated and dispositioned according to the applicable procedures for handling of nonconforming material.

5.15 INSPECTION OF NONFLYING HARDWARE

Unless otherwise specified nonflying hardware shall be inspected as follows:

5.15.1 Requirements Agreement

It is mandatory the agreement between the test proposer and the Embraer Level 3 to establish the specific requirements that may apply to the objectives of the referred inspection.

5.15.2 Inspection for Material Homogeneity

When nonflying hardware is to be inspected for material homogeneity only, the inspection process shall be capable of detecting localized discontinuities like delamination, inclusions and voids. It shall also be capable of qualitatively detecting porosity; however assessment of porosity content based on signal attenuation is optional.

This type of inspection is most frequently applicable to sample panels used for:

- Initial characterization of materials (screening of materials);
- Materials qualification;
- Initial verification of manufacturability; and
- Material certification campaigns (design allowables).

Reference standards according to 5.8 are not required for this type of inspection. Equipment standardization may be accomplished on the hardware itself.

When deemed necessary by the Embraer Level 3, appropriate reference standards shall be used to ensure sensitivity and comparison among inspections. The applicable reference standards shall comply at least with the following:

- Laminate materials shall be of the same type as those of the applicable hardware;
- Laminate thickness shall be within $\pm 0,5$ mm of the laminate thickness of the applicable hardware; and
- When deemed necessary the porosity content shall be verified and within acceptance limits applicable to the material of the hardware to be inspected.

Attenuative materials like lead tape cutouts may be attached to the surface of applicable hardware to ensure adequate sensitivity.

Unless otherwise required by the Embraer Level 3, this type of inspection may be performed by inspector ongoing job training under certified personnel assistance.

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The inspection report content and details shall be as required according to inspection objectives.

All other aspects of the applicable inspection process shall be as approved by the Embraer Level 3.

5.15.3 Inspection for Production Process Assessment

When nonflying hardware is to be inspected for production process assessment, the inspection process shall be compliant to all applicable requirements of this document with exemptions herein specified.

This type of inspection is most frequently applicable to aircraft parts or subcomponents commonly subjected to tear down after NDT inspection such as:

- Assessment of product manufacturability;
- Demonstrators within the scope of R&D projects; and
- Production try-outs of new products or product changes.

This type of inspection does not apply to FPQ and FAI parts.

Equipment standardization shall be performed on PSRS as specified by the Embraer Level 3 and shall be compliant to the requirements of 5.8.6.

Unless otherwise approved by the Embraer Level 3, this type of inspection shall be performed by certified personnel.

The inspection report content and details shall be as required according to inspection objectives.

6. QUALITY ASSURANCE PROVISIONS

6.1 QUALIFICATION OF INSPECTION PROCESSES

6.1.1 The inspection process, including applicable reference standards and written procedure, shall be qualified as follows:

- a) the inspection process shall comply with all applicable requirements; and
- b) the effectiveness of the inspection process shall be demonstrated on a part undergoing first part qualification (FPQ) tests in accordance with the acceptance criteria defined in material specification (MEP) and NE 20-032:

- when ultrasonic inspection techniques are used, compliance to the requirements of 5.9.6 and 5.9.7 shall be achieved;
- acceptable signal to noise ratio shall be verified;
- sensitivity setting as per 5.9.2 shall be demonstrated;
- detection of indications as per 5.11.5 shall be demonstrated;

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- detection of RSD located in part features as per 5.7.6 shall be verified; and
- the results of FPQ tests shall comply with requirements.

6.1.2 The qualified inspection process and applicable reference standards may be used for acceptance inspection of all parts included in the approved FPQ.

6.1.3 Unless otherwise specified, the Embraer Level 3 shall witness the qualification of the inspection process.

7 GENERAL NOTES

7.1 DRAWING CALLOUT

This Standard shall be called out on Part List of Engineering Drawings through:

- NI 360 - Nondestructive inspection according to NE 07-011.

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